Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number

Learner Registration Number

Pearson BTEC Level 3 Nationals Certificate, Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Time 40 minutes

Paper reference **31617H/1P**

Applied Science/Forensic and Criminal Investigation

UNIT 1: Principles and Applications of Science I

Physics

SECTION C: WAVES IN COMMUNICATION

You must have:
A calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided – *there may be more space than you need.*

Information

- The exam comprises three papers worth 30 marks each:
  – Section A: Structures and Functions of Cells and Tissues (Biology)
  – Section B: Periodicity and Properties of Elements (Chemistry)
  – Section C: Waves in Communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- The formulae sheet can be found at the back of this paper.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

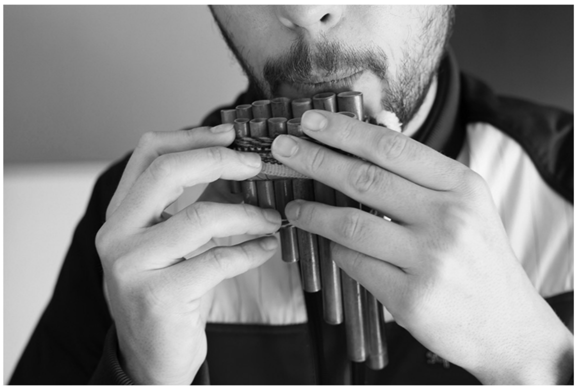
Turn over ▶

Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 (a) Musical sounds of different frequencies can be made by blowing across pipes.

Figure 1 shows a person blowing across a set of pipes.



(Source: © The Print Collector/Alamy Stock Photo)

Figure 1

Figure 2 shows a stationary wave produced by blowing into an open-ended pipe.

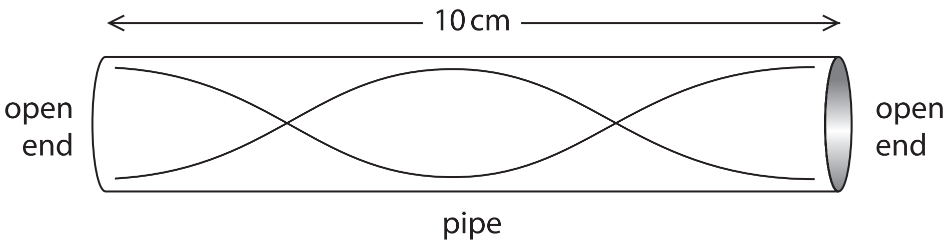


Figure 2

(i) Identify the wavelength of the stationary wave in the pipe. **(1)**

☒ **A** 5 cm

☒ **B** 10 cm

☒ **C** 15 cm

☒ **D** 20 cm

(ii) Add an X to Figure 2, to show the position of an antinode. **(1)**

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P 7 4 5 5 0 A 0 2 1 2

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(b) The frequency (f) of a sound wave is 1370 Hz.

The velocity (v) of the sound is 330 ms^{-1}

Calculate the wavelength (λ) of the wave.

Use the equation $v = f \times \lambda$.

Show your working.

(3)

wavelength (λ) = ... m

(c) Give **two** ways that the frequency of the sound wave in Figure 2 can be changed.

(2)

1 ....

2 ....

(Total for Question 1 = 7 marks)

2 (a) Digital signals are used for communication.

Draw a digital signal in Figure 3.

(1)

Figure 3

(b) Broadband is a system that gives rapid internet access.

Coaxial cables can be used to send and receive broadband signals.

(i) Give **two other** methods that can be used to send and receive broadband signals.

(2)

1 ....

2 ....

(ii) Broadband is a system that allows many signals to be transmitted at the same time.

Complete Sentence 1 to show how signals are transmitted by broadband.

(1)

Broadband sends many signals at the same time by a process called ....

Sentence 1

(Total for Question 2 = 4 marks)

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3 (a) Figure 4 shows a ray of light at an air–glass interface.

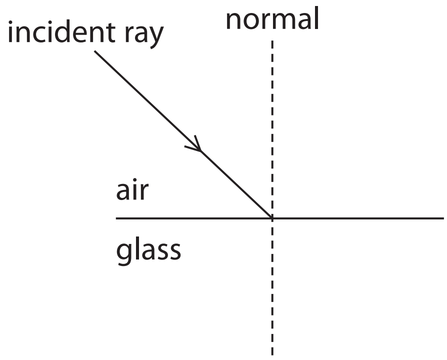

Figure 4

When light passes from **air** into **glass** the light changes speed and direction.

Identify the correct change as the ray of light enters the glass.

(1)

		change in speed	change in direction
☒	**A**	is faster in glass	moves towards the normal
☒	**B**	is faster in glass	moves away from the normal
☒	**C**	is slower in glass	moves towards the normal
☒	**D**	is slower in glass	moves away from the normal

(b) Figure 5 shows a ray of light at a glass–air interface.

The ray of light strikes the surface of the glass at the critical angle *c*.

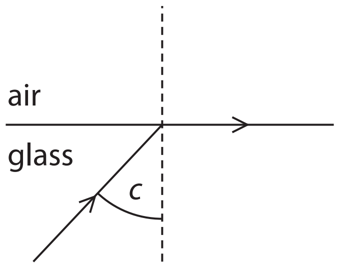

Figure 5

The refractive index (*n*) of the glass is 1.52

Show that the critical angle *c* is approximately 41°

(4)

Use the equation

$$n = \frac{1}{\sin c}$$

critical angle *c* .... °

(c) Figure 6a shows light incident at 45° at a glass–water interface.

Figure 6b shows light incident at 45° at a glass–air interface.

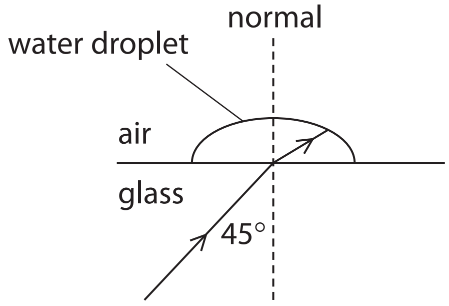

Figure 6a

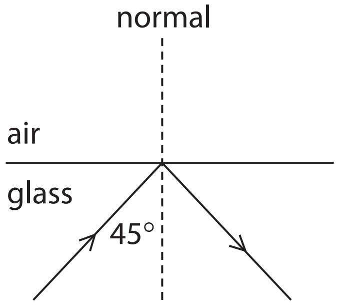

Figure 6b

The critical angle for a glass–water interface is 62°
The critical angle for a glass–air interface is 41°

Explain why light is **refracted** in Figure 6a but light is **reflected** in Figure 6b.

(3)

....

....

....

....

....

....

(Total for Question 3 = 8 marks)

4 (a) Which statement describes all electromagnetic waves?

(1)

- ☒ **A** longitudinal with different frequencies
- ☒ **B** longitudinal with the same frequency
- ☒ **C** transverse with different frequencies
- ☒ **D** transverse with the same frequency

(b) Mobile phone networks use tower masts as base stations.

Base stations receive signals from mobile phones.

Explain why a large city needs to have many base stations.

(2)

....

....

....

....

(c) Bluetooth© is a system that is used to connect electronic devices.

Explain why Bluetooth© signals do not need 'line of sight' to communicate with other Bluetooth© devices.

(2)

....

....

....

....

(Total for Question 4 = 5 marks)

5 Figure 7 shows an example of an emission spectrum of a gas.

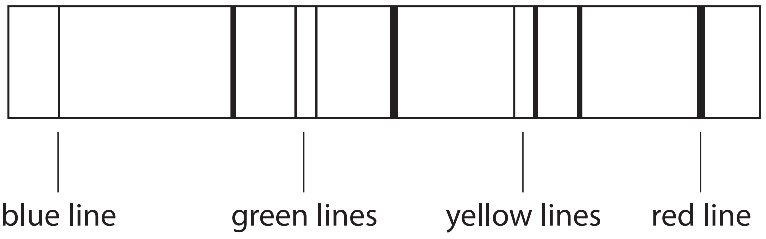

Figure 7

Discuss how emission spectra are produced and can be used to identify the elements in a gas.

(6)

(Total for Question 5 = 6 marks)

TOTAL FOR SECTION C = 30 MARKS
TOTAL FOR EXAM = 90 MARKS

Formulae sheet

Wave speed $v = f\lambda$

Speed of a transverse wave on a string $v = \sqrt{\dfrac{T}{\mu}}$

Refractive index $n = \dfrac{c}{v} = \dfrac{\sin i}{\sin r}$

Critical angle $\sin C = \dfrac{1}{n}$

Inverse square law in relation to the intensity of a wave $I = \dfrac{k}{r^2}$

$$\frac{I_1}{I_2} = \frac{(D_2)^2}{(D_1)^2}$$

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